

What is Apache Kafka?

- **Apache Kafka** is a **distributed event streaming platform**.
- Used for:
 - Real-time data pipelines
 - Streaming analytics
 - Event-driven systems

👉 Simple:

Kafka = **publish** + **subscribe** messaging system (like WhatsApp group for services)

Real-World Example: Food Delivery App 🍔

Flow:

1. User places order
2. Order Service → sends event to Kafka (topic)
3. Multiple services consume:
 - Payment Service 💳
 - Restaurant Service 🍷
 - Delivery Service 🚲

👉 One message → multiple consumers

👉 Services are **decoupled (independent)**

Why Use Kafka?

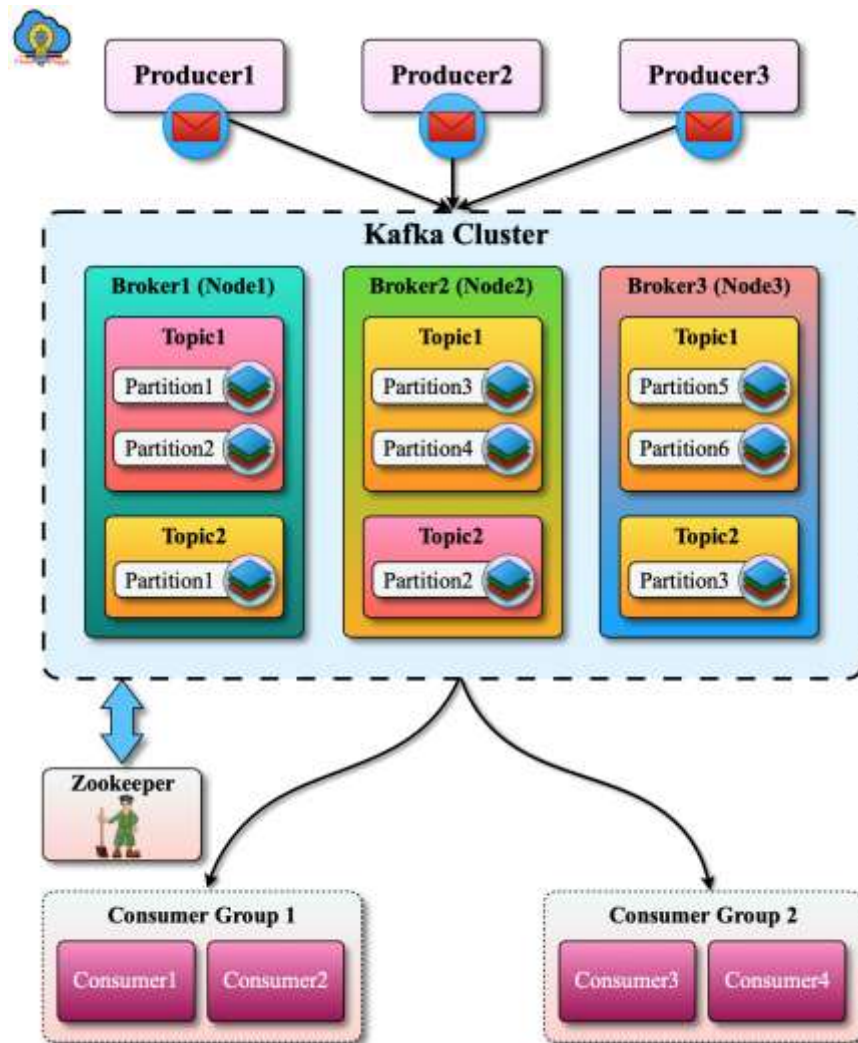
- High Throughput (handles millions of messages)
- Fault Tolerance (replication)
- Durable (data stored on disk)
- Scalable (add more brokers easily)

Architecture

Components:

- **Producer**
→ Sends data to Kafka
- **Consumer**
→ Reads data from Kafka
- **Broker**
→ Kafka server (stores data)

- **Topic**
→ Category of messages (like orders, payments)
- **Partition**
→ Split of topic for parallel processing
- **ZooKeeper**
→ Manages brokers (leader election, metadata)



How to Run Kafka (Windows)

1. Start ZooKeeper

```
bin\windows\zookeeper-server-start.bat config\zookeeper.properties
```

2. Start Kafka Server

```
bin\windows\kafka-server-start.bat config\server.properties
```

3. Create Topic

```
bin\windows\kafka-topics.bat --create --topic topic1 --bootstrap-server  
localhost:9092 --partitions 4 --replication-factor 1
```

4. List Topics

```
bin\windows\kafka-topics.bat --list --bootstrap-server localhost:9092
```

5. Describe Topic

```
bin\windows\kafka-topics.bat --describe --topic topic1 --bootstrap-server  
localhost:9092
```